Predicting Snow Depth in Mountain Terrain

Empirical Bayesian Kriging with and without terrain covariates in Washington State

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Method Selection and Justification

Snow depth in Washington State is highly variable due to complex terrain, including elevation gradients, slope, aspect, and vegetation cover. To model this spatial variability and associated uncertainty, Empirical Bayesian Kriging (EBK) and Empirical Bayesian Kriging Regression (EBK Regression) were applied. EBK was selected as a baseline interpolation method because it accounts for spatial autocorrelation while automatically estimating variogram parameters through repeated simulations, reducing subjectivity and improving robustness for heterogeneous environmental data.

EBK Regression was implemented to extend the baseline model by incorporating terrain covariates elevation, slope, aspect, and canopy cover which are known to influence snow accumulation. Including these variables allows the model to explain variation not captured by spatial proximity alone. Comparing EBK and EBK Regression provides insight into whether terrain controls meaningfully improve prediction accuracy and reduce uncertainty, making this paired approach well suited to snow depth modeling in mountainous regions.

Interpretation of Spatial Patterns

Emperical Bayesian Kriging Standard Error vs Emperical Bayesian Kriging Prediction

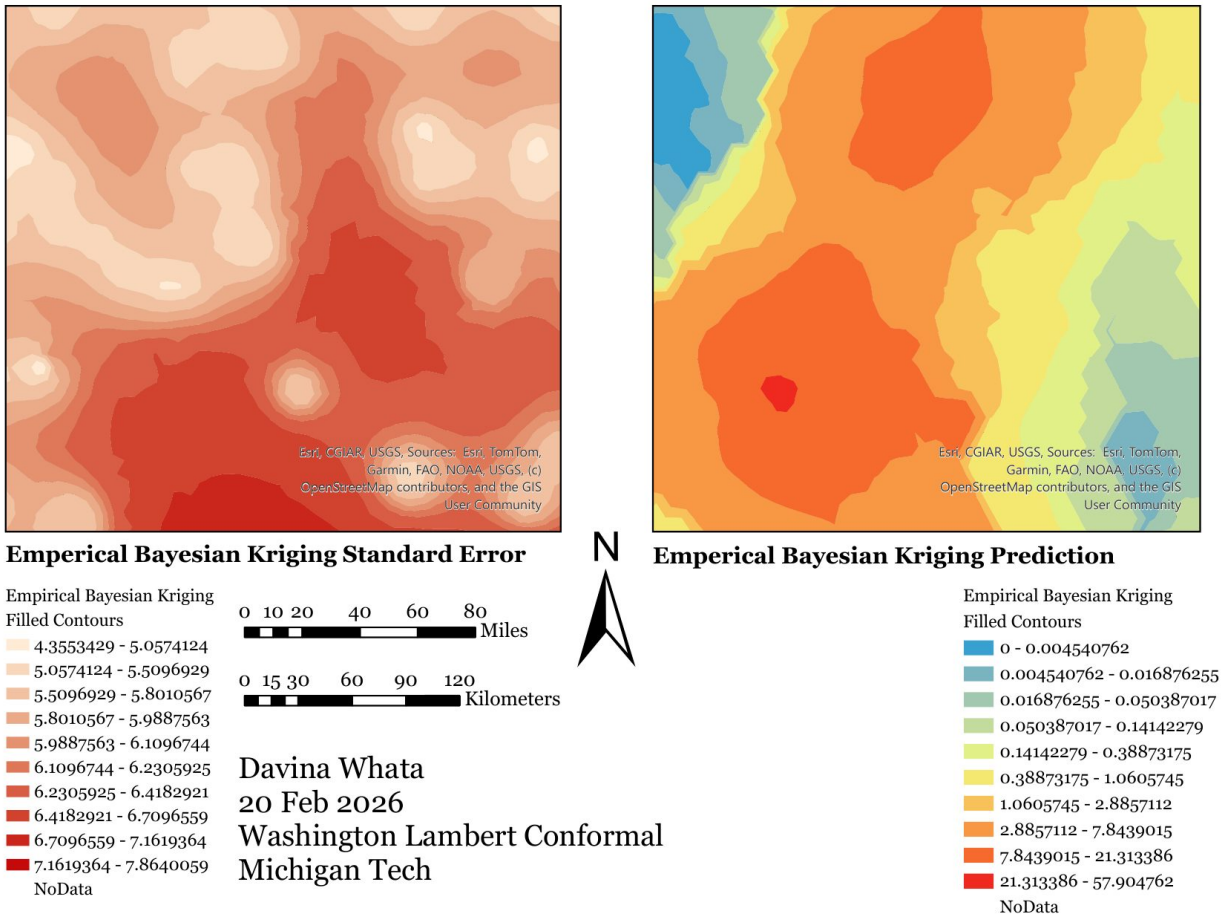


Figure 1. Empirical Bayesian Kriging: standard error (left) and predicted snow depth (right).

The EBK prediction surface displays smooth, generalized patterns of snow depth across the study area. Higher predicted values are concentrated in interior and higher-elevation regions, while lower values occur toward peripheral and lowland areas. These patterns primarily reflect the influence of nearby sample points and spatial autocorrelation rather than explicit environmental drivers.

EBK Regression Standard Error vs EBK Regression Prediction

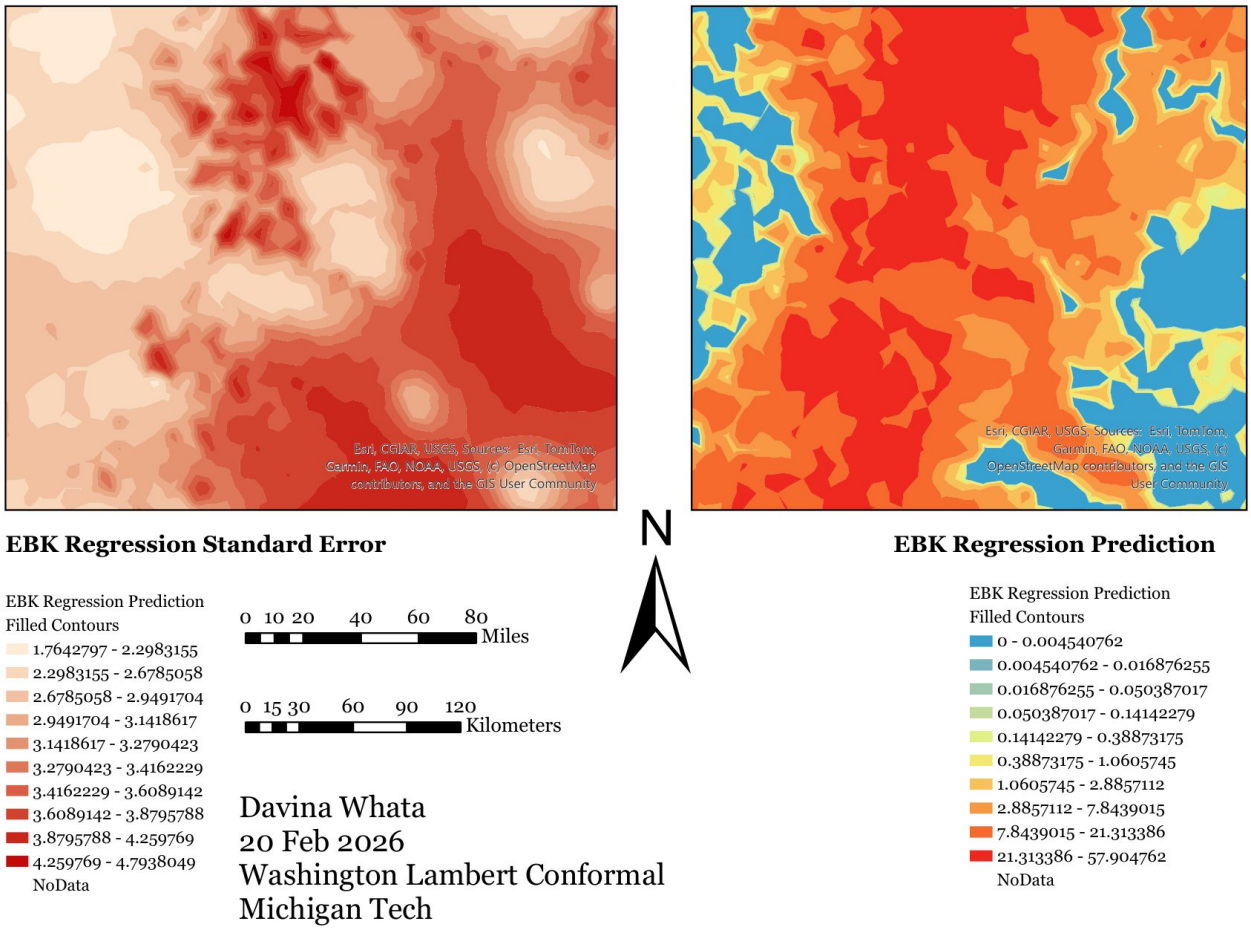


Figure 2. EBK Regression with terrain covariates: standard error (left) and predicted snow depth (right).

In contrast, the EBK Regression prediction map exhibits more spatial detail and sharper transitions. Areas of higher snow depth align more closely with terrain-controlled regions, suggesting that elevation and related covariates play a significant role in shaping snow distribution. This indicates that the regression model captures physical processes influencing snow accumulation that are not fully represented in the EBK-only model.

Uncertainty and Limitations

Uncertainty patterns differ notably between the two models. The EBK standard error surface shows extensive areas of elevated uncertainty, particularly in locations far from sampling points. These red and darker zones indicate where predictions are least reliable due to limited local data support.

The EBK Regression standard error map shows reduced uncertainty across much of the study area, especially in topographically complex regions. This reduction demonstrates that incorporating terrain variables improves prediction confidence where elevation and slope help explain snow depth variability. However, areas with persistently high standard error remain, indicating locations where terrain variables alone are insufficient or where sampling density is particularly low.

Key limitations include reliance on the assumption of spatial stationarity and the quality of covariate datasets. Additionally, while EBK Regression improves uncertainty estimates, it cannot fully compensate for sparse sampling. Increasing field measurements in high-uncertainty areas would further improve model performance.

Conclusion

Overall, EBK Regression outperforms EBK by producing more realistic spatial patterns and lower prediction uncertainty. However, both models highlight the critical influence of sampling density and reinforce the importance of interpreting prediction maps alongside standard error surfaces.